

The effect of acute pre-exercise dark chocolate consumption on plasma antioxidant status, oxidative stress and immunoendocrine responses to prolong...

PURPOSE: Acute antioxidant supplementation may modulate oxidative stress and some immune perturbations that typically occur following prolonged exercise. The aims of the present study were to examine the effects of acutely consuming dark chocolate (high polyphenol content) on plasma antioxidant capacity, markers of oxidative stress and immunoendocrine responses to prolonged exercise. **METHODS:** Fourteen healthy men cycled for 2.5 h at ~60% maximal oxygen uptake 2 h after consuming 100 g dark chocolate (DC), an isomacronutrient control bar (CC) or neither (BL) in a randomised-counterbalanced design. **RESULTS:** DC enhanced pre-exercise antioxidant status ($P = 0.003$) and reduced by trend ($P = 0.088$) 1 h post-exercise plasma free [F $\square$ -isoprostane] compared with CC (also, [F $\square$ -isoprostane] increased post-exercise in CC and BL but not DC trials). Plasma insulin concentration was significantly higher pre-exercise ($P = 0.012$) and 1 h post-exercise ($P = 0.026$) in the DC compared with the CC trial. There was a better maintenance of plasma glucose concentration on the DC trial (2-way ANOVA trial $\times$ time interaction $P = 0.001$), which decreased post-exercise in all trials but was significantly higher 1 h post-exercise ($P = 0.039$) in the DC trial. There were no between trial differences in the temporal responses (trial $\times$ time interactions all $P > 0.05$) of hypothalamic-pituitary-adrenal axis stress hormones, plasma interleukin-6, the magnitude of leukocytosis and neutrophilia and changes in neutrophil function. **CONCLUSION:** Acute DC consumption may affect insulin, glucose, antioxidant status and oxidative stress responses, but has minimal effects on immunoendocrine responses, to prolonged exercise.